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Methods and system for controlling access to enterprise resources based on tracking

Abstract

Methods and systems for controlling access to enterprise resources based on tracking are disclosed. In one implementation, an enterprise security system includes one or more tracking systems, an information technology (IT) system, and a security integration system (SIS). The one or more tracking systems track movement of individuals throughout the enterprise. The SIS blocks access to the computer resources based on the tracked movement of the individuals within the enterprise.

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Background/Summary

RELATED APPLICATIONS (1) This application is a § 371 National Phase Application of International Application No. PCT/US2018/058971, filed on Nov. 2, 2018, now International Publication No. WO 2019/090087, published on May 9, 2019, which International Application claims the benefit under 35 USC 119 (e) of U.S. Provisional Application No. 62/581,207, filed on Nov. 3, 2017, both of which are incorporated herein by reference in their entirety.

BACKGROUND OF THE INVENTION

(1) Enterprises, such as private and public companies, municipal, state and federal governmental agencies, and other entities, will often maintain a number of disparate systems to support their operations, track their business relationships, and maintain security. Enterprise Resource Planning (ERP) systems are computer systems that allow enterprises to manage operations. Employee Resource Management (ERM) systems are typically different computer systems that allow the enterprises to track, schedule, and pay their employees. Access control systems are principally concerned with physical security and the selective access to, restriction of access to, and/or notification of access to the enterprises' buildings and/or secured parts of those buildings. Information technology (IT) security systems are principally concerned with control and restricting access to computer resources, often including access to any ERP or ERM systems, and possibly other systems such as file servers, etc. In addition, other security systems are often employed by the enterprises to round-out their security needs. A common example is a surveillance system, which may be integrated with any access control system to varying degrees.

(2) The ERM systems store and manage many different types of information associated with employees. The ERM system might execute on a single computer system or server, or across multiple computer systems and servers, or be implemented in a cloud-based computer system. The different types of employee information often controlled and managed by the ERM systems include biographic, including demographic, information, payroll and salary information, job performance and attendance information, benefits information, and training and compliance information, to list some common examples.

(3) Modern ERM systems typically encompass the functionality of multiple legacy systems that had separately managed and stored the different types of information associated with the employees. These legacy systems might have had separate payroll systems for the payroll and salary information, human resources systems for the biographic, job performance and attendance

information, benefits systems for the benefits information, and learning systems for the training and compliance information, in examples. At the same time, the ERM system can simply be a collection of local or remote databases that store the different types of information associated with each employee.

(4) ERP systems generally track business resources such as cash, raw materials, and production capacity. The ERP systems can track business commitments such as orders, purchase orders, and possibly might overlap with ERM systems. As such, these ERP systems are computer systems that often include computer databases and database management systems for the databases, along with different user interfaces. The different user interfaces allow the operators to obtain information from the systems and submit information to them. The ERP systems typically further include various computer systems connected with manufacturing and the control and management of other operations of the enterprises.

(5) IT security systems can take many forms. They will often include asset management database computer systems that allow the enterprises to track their computers, the status of the computers, individuals to whom the computers are assigned, along with other systems such as servers, cloud server systems, and networking devices, to list some examples. The IT security systems will almost always include one or more authentication server systems. These computers provide network services that computer systems and applications that execute on those computer systems use to authenticate the credentials, usually account names and passwords, of their users. For example, when a client computer successfully submits a valid set of credentials that were entered by a user trying to log onto the computer, the client computer receives a cryptographic ticket that it can subsequently use to access various other systems such as the enterprise's ERP systems, ERM systems and other computers systems such as even mundane file servers.

(6) The access control systems typically include access control readers. These readers are often installed at access points of the buildings to control access to restricted areas, such as buildings or areas of the buildings. Examples of access points include front and interior doors of a building, elevators, hallways connecting two areas of a building, to list a few examples. The access control readers authenticate identities of (or authorize) individuals and then permit those authenticated individuals to access the restricted areas through the access points. Typically, individuals interact with the access control readers by swiping keycards or bringing contactless smart cards within range (approximately 2-3 inches or 5 centimeters) of a reader. The access control readers read user information of the keycards, such as credentials of the individuals, and then the access control systems determine if the individuals are authorized to access the restricted areas. If the individuals are authorized to enter the restricted areas, then the access control readers allow access to the restricted areas by unlocking locked doors, signaling that doors should be unlocked, or generating alarms upon unauthorized entry, for example.

(7) More recently, frictionless access control systems are being proposed and designed. These systems typically rely on individuals carrying beacon devices that can broadcast credentials, such as dedicated fob devices or personal mobile computing devices such as tablet or smart phone computing devices. These systems are “frictionless” in that the individual may not have made any overt gesture indicating a desire to access the restricted area, e.g., the individuals did not swipe a keycard. The access control systems will then monitor and track the individuals as they move through the buildings and automatically open access points such as doors when approached, assuming that the individuals are authorized to pass through those access points.

(8) Enterprise surveillance systems are used to help protect people, property, and reduce crime. These systems are used to monitor buildings, lobbies, entries/exits, and secure areas within the buildings of the enterprises, to list a few examples. The surveillance systems record illegal activity such as theft or trespassing, in examples. At the same time, these surveillance systems can also have business uses. They can track employee locations across different rooms within buildings and among the different buildings of the enterprises.

(9) In these surveillance systems, surveillance cameras capture image data of scenes. The image data is typically represented as two-dimensional arrays of pixels. The cameras then include the image data within streams, and users of the system such as security personnel view the streams on display devices such as video monitors. The image data is also typically stored to a video management system (VMS) for later access and analysis.

(10) Increasingly, it is being proposed to make these VMSs smarter. For example, VMSs with image analytics systems are becoming more prevalent. These analytics systems allow the VMSs to interpret the captured image data and possibly send alerts when detecting trespassing or other problems. One common type of image analytics is facial recognition. In such instances, modules of the VMS will identify faces within the image data and compare those faces to databases to try to identify the different individuals captured in the image data.

SUMMARY OF THE INVENTION

(11) A proposed enterprise security system obtains information from disparate repositories and combines the information with tracking information to validate movement of individuals within the enterprise and also uses the information to validate log-on events on computer systems. In one application, the enterprise security system uses the combined information to determine whether the individuals are located in and/or moving to a correct/authorized location for those individuals. For example, based upon a desk location, it will be possible to monitor the location of an individual within a building and determine that they are in and/or are moving to the correct location within the building.

(12) Additional functions are possible. The system can use biographic and physical information such as age, gender, facial hair, and whether or not the person is wearing glasses, in examples, to determine that obtained facial recognition information matches that of an employee record maintained by the ERM system. The system can combine facial recognition with information technology (IT) systems monitoring to confirm that a person whose face has been recognized uses the correct login identifier to log into a computer system within the building(s) of the enterprise.

(13) In general, according to one aspect, the invention features an enterprise security system. The system includes one more tracking systems, an IT system and a security integration system. The one or more tracking systems track movement of the individuals throughout the enterprise. The IT system provides access to computer resources within the enterprise. The security integration system can then generate commands to block access to the computer resources based on the tracked movement of the individuals within the enterprise.

(14) In this way, the system might augment the validation provided by a traditional authentication server by ensuring that log-on attempts are consistent with the individuals actually present in a conference room, for example.

(15) In embodiments, the tracking systems include a surveillance system including surveillance cameras and a facial recognition module for performing facial recognition of individuals in the image data from the surveillance cameras. The tracking systems can also, or alternatively, include access control systems that control access of individuals through access points within the enterprise.

(16) Typically, the security integration system determines authorized computer systems for individuals and instructs the IT system to block access to the computer systems when authorized users of the computer systems are not present. The security integration system can also access an asset management database to determine locations of the computer systems within the enterprise.

(17) In general, according to another aspect, the invention features an enterprise security system. The enterprise security system includes one or more tracking systems for tracking movement of individuals throughout an enterprise, an employee resource management system including an employee database, and a security integration system. The employee database stores location of desks of the employees. The security integration system receives a metadata stream of the tracked movement of the employees and determines whether the employees are moving to and/or from the

respective desks of the employees by reference to the information from the employee database.

(18) In general, according to yet another aspect, the invention features an enterprise security method. The method includes tracking movement of individuals throughout an enterprise, providing access to computer resources within the enterprise using an authentication server, and blocking access to the computer resources based on the tracked movement of the individuals within the enterprise.

(19) In one example, tracking movement includes controlling and reporting access of individuals through access points within the enterprise. Tracking movement can further or alternatively include capturing images of the individuals and performing facial recognition of the individuals.

(20) In general, according to still another aspect, the invention features another enterprise security method. This method includes tracking movement of individuals throughout an enterprise, an employee resource management system including an employee database indicating location of desks of the employees, and receiving a metadata stream of the tracked movement of the employees and determining whether the employees are moving to and/or from the respective desks of the employees.

(21) The above and other features of the invention including various novel details of construction and combinations of parts, and other advantages, will now be more particularly described with reference to the accompanying drawings and pointed out in the claims. It will be understood that the particular method and device embodying the invention are shown by way of illustration and not as a limitation of the invention. The principles and features of this invention may be employed in various and numerous embodiments without departing from the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) In the accompanying drawings, reference characters refer to the same parts throughout the different views. The drawings are not necessarily to scale; emphasis has instead been placed upon illustrating the principles of the invention. Of the drawings:
- (2) FIG. 1 is a schematic diagram showing rooms of a building and an enterprise security system installed at the building, in accordance with principles of the present invention;
- (3) FIG. 2A is a sequence diagram that illustrates one method of operation for the enterprise security system in FIG. 1, where the method determines whether an otherwise authorized individual, at a computer system in a room within the building, is authorized to log into a computer system within the building;
- (4) FIG. 2B is a sequence diagram showing another embodiment where the enterprise security system generates a black list; and
- (5) FIG. 3 is a sequence diagram that illustrates another method of operation for the enterprise security system, where the system determines whether an otherwise authorized individual is traversing a path near desk locations to which they are assigned.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (6) The invention now will be described more fully hereinafter with reference to the accompanying drawings, in which illustrative embodiments of the invention are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art.
- (7) As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Further, the singular forms and the articles “a”, “an” and “the” are intended to include the plural forms as well, unless expressly stated otherwise. It will be further understood that the terms: includes, comprises, including and/or comprising, when used in this specification,

specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. Further, it will be understood that when an element, including component or subsystem, is referred to and/or shown as being connected or coupled to another element, it can be directly connected or coupled to the other element or intervening elements may be present.

(8) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(9) FIG. 1 shows an exemplary enterprise including an enterprise security system **100**, which has been constructed according to the principles of the present invention.

(10) The figure shows client computer systems **122** at a building **50** of the enterprise. Some of these client computer systems might be public terminals in conference rooms that might be used by different individuals over the course of the day. In other examples, the client computer systems **122** are dedicated computer systems assigned to a specific employees of the enterprise or shared computers systems such as servers or server systems.

(11) Also shown are other computer systems that are associated with the operations of the enterprise, such an enterprise information technology (IT) system **20**, a surveillance system, an ERM system **138**, a file server **42**, an access control system **128**, and an ERP system **44**. Finally, a security integration system (SIS) **80** provides some of the important functions of the present invention. A local area or enterprise network **210** provides network connectivity.

(12) In more detail, the cameras **103** of the surveillance system capture image data **99** of the rooms **113** throughout the enterprise's building and thus images of individuals **60** in each room **113**. Cameras **103-1** and **103-2** are respectively installed in rooms **113-1** and **113-2**.

(13) The surveillance system's VMS **110** stores the image data **99** from the cameras **103** and includes a camera locations table **21** and a facial recognition module **107**. The camera locations table **21** typically has a record for each of the surveillance cameras **103**. The record contains such information as the room in which the camera is installed. It may also include information concerning the type of camera and possibly even the field of view of the camera with respect to a map or floor layout of the building **50**. The facial recognition module **107** determines facial recognition information of the individuals captured in the image data and monitors movement and/or activity of individuals **60** within the rooms **113**.

(14) The IT system **20** includes an authentication server **22**. The authentication server **22** allows the individuals **60** to log onto the computer systems **122**, such as client computer systems, and then typically access other resources such as the file server **42** and/or the ERP system **44** and/or ERM system **138**. For this purpose, the authentication server **22** has a user database **39** that stores user records **149** for the individuals **60**. The individuals **60** are typically employees of the enterprise. Each user record **149** includes information that identifies each individual and the computer systems **122** each is authorized to access, in examples. In more detail, the user record **149** for each individual **60** typically includes a user ID **72**, a login ID **144**, user credentials including passwords **172**, and a list of authorized computer systems **179**.

(15) Another common feature of IT systems is an asset management database **30**. The asset management database **30** includes a computer locations table **96**. As its name implies, the table **96** maintains locations (e.g. room numbers) for each of the computer systems **122**. Often, these asset management databases **30** will include additional information such as specific details concerning the computer systems **122**, such as production year operating system, applications installed on the computer systems and possibly other license information.

(16) The ERM system **138** is preferably as described hereinabove. As such, it has an employee database **139** that stores employee records **123** of employees **60**. The employee records **123** include information for identifying each employee and locations of desks **45** within the building **50** for the employees. In more detail, each employee record **123** typically includes a name **24**, an employee number **32**, a badge number **34**, a badge photo **36**, user biographic and physical information **141**, an authorization level **151**, and one or more desk locations **161**. The desk locations **161** list the locations of desks that each employee is authorized to be present at or is otherwise expected to be near during work hours. In addition, the ERM system **138** may also include other information such as databases that store the same information for contractors and visitors to the enterprise.

(17) The user biographic and physical information **141** includes other information for possibly identifying each individual **60**. This information might include the following: a birthplace, home address, age, race, hair color/length, and whether the individual typically wears glasses, in examples.

(18) The access control system **128** controls physical access through access points **10** of the building **50**. In the illustrated example, the access points are doors, but may also include hallways or elevators or floors within the buildings of the enterprise. Typically, the access control system **128** further includes card readers for reading employee badges and/or frictionless readers that might validate employees based on credentials provided by a mobile computing device such as a smart phone operated by the employees. In this way, the access control system is able to monitor movement of individuals through access points.

(19) In one embodiment, the enterprise security system includes one or more tracking systems (such as the surveillance system and/or the access control system). These one or more tracking systems track movement of the individuals **60** throughout the enterprise and provide this information as meta data streams. The IT system **20** provides access to computer resources within the enterprise and provides its own meta data stream of log-on events.

(20) The SIS **80** functions to integrate the operation of the tracking system(s) and the IT system. As such, the SIS **80** can be a separate computer system or could be a process or module that executes on the IT system computers, e.g., authentication server, or on a computer associated with the tracking systems, or even a separate computer system or a computer system integrated with the ERP or ERM computer systems. In any case, the SIS instructs the IT system **20** to block access to the computer systems **122** when authorized users of the computer systems **122** are not present.

(21) In the illustrated example, the computer systems **122** are located in the rooms **113**. Room **111-1** includes computer system **122-1**. Room **113-2** includes computer system **122-2** upon a desk **45**. The locations and types of these computers are stored in the computer locations table **96** of the asset management database **30**. Multiple individuals **60** are located within and/or are moving about the rooms **113**, and are attempting to access computer resources from within the rooms **113**. Individual **60-1** is located in room **113-1** and individual **60-2** is located in room **113-2**. Individual **60-2** is also located near the desk **45** and is attempting to access/log onto a user account via the computer system **122**.

(22) In general, the individuals **60** might be employees. As such, when the individuals are originally hired as employees, a guard or security operator or human resources representative would create the employee record **123** for each employee in the employee database **139**. The security operator also takes a picture of the employee's face to use as the badge photo **36**, and additionally uses the facial recognition module **107** of the VMS **110** to create stored facial recognition information for each of the employees.

(23) The facial recognition information can be of different types. In one example, the information is a biometric identifier such as a facial signature of the individual. In another example, the information is simply a still image of the person's face extracted from the image data, also known as a facial patch.

(24) The facial signature for an individual is a unique value or set of values that represent the face

of an individual/employee. The facial recognition module **107** uses one or various predetermined facial signature algorithms to create the facial signature, based upon various features of each person's face. These features include the eyes, nose, mouth, eyebrows, cheekbones, and chin of each face, and distances between each of these features, in examples.

(25) The facial recognition module also maps each instance of facial recognition information (e.g. the facial signature or facial patch) for each employee to a user credential or other identifier (OD). In this way, the OD associated with each instance of stored facial recognition information can be used to identify the individual for which the facial signature was obtained.

(26) The VMS **110** then stores the facial recognition information and associated ID for identifying each employee. In one example, the VMS stores this information locally to the VMS **110**. In another example, the VMS **110** might store this information to the employee record **123** for each employee.

(27) The enterprise security system **100** generally operates as follows. After the individuals **60** are registered as employees, one or more individuals arrive at access points **10** such as doors of the building **50**. The individuals typically present keycards that include their user credentials to the card readers **70** of the access control system **128** at the doors. The cameras **103** also capture and send the image data **99** of the individuals to the VMS **110**.

(28) The VMS **110** then uses its facial recognition module **107** to determine whether the images of the individuals **60** in the image data **99** from the surveillance cameras match the stored facial recognition information for registered employees. If the individuals **60** are determined to be employees, the VMS **110** operates in conjunction with the SIS **80**, the IT system **20**, the ERM **138**, and possibly the ERP system **44** to determine whether the individuals are: in the correct rooms within the building **50**; attempting to access computer systems to which they are authorized; and traversing paths near desk locations to which they are assigned, in examples.

(29) FIG. 2A is a sequence diagram that illustrates one application of the enterprise security system **100**.

(30) The enterprise security system **100** protects against improper access of computer systems **122** within the rooms **113** of the building. After the identity of the individual **60** entering or within a room **113** has been determined via facial recognition, only IT user records **149** associated with that individual will be allowed to be accessed from computer systems located within that room **113**. Thus, if an otherwise authorized user **60** enters the room **113** and tries to use another person's credentials, the individual **60** will be prohibited from doing so. In this way, only an individual **60** that is expected to be near a computing system **122** at a specific location can log onto that computer system **122**, in one example.

(31) In more detail, in step **202**, one or more card readers **70** read keycard credentials of individuals **60** presented at the card readers. The card readers **70** send the credentials to the ACS **128** in step **204**.

(32) According to step **206**, the ACS **128** sends an indication as whether the individuals **60** are authorized. In one example, the indication is a door lock signal to unlock or otherwise disengage the door locks for the doors (or other access points) at which the card readers **70** are installed. The ACS **128** sends these indications to the door controller **112**. Then, in step **208**, the ACS **128** reports these credentials of the individuals **60** and the room **113** that each entered to the SIS **80** as an access control meta data stream.

(33) At the same time, in step **210**, the cameras **103-1** and **103-2** within the rooms **113-1** and **113-2** of the building **50** capture the image data **99** of scenes in the rooms **113**. The cameras **103** send the image **99** to the VMS **110** for storage and subsequent analysis.

(34) In step **212**, the facial recognition module **107** of the VMS **110** locates the individuals **60** in the image data **99**. In step **214**, the facial recognition module **107** then performs facial recognition of the individuals located in the image data **99** to identify the individuals **60**. For this purpose, the facial recognition module **107** preferably uses the same facial recognition algorithms used when the

security operators first registered the individuals as employees. In this way, the facial recognition module **107** can identify employees in the image data by reference to the badge photos **36** stored in the employee database **139** of the ERM system **138**.

(35) It can also be appreciated that the facial recognition module **107** can be included within and execute upon other components of the enterprise management system **100**. In one example, the facial recognition module **107** might be integrated within the cameras **103** and execute upon a microcontroller of the cameras **103**. In another example, the facial recognition module **107** might execute upon a microcontroller or central processing unit (CPU) of the ACS **128**.

(36) In step **216**, the VMS **110** provides a surveillance meta data stream of the identified individuals, using facial recognition, and the rooms **113** in which those individuals are located based on the location of the various cameras **103** that captured their image, along with a timestamp indicating the time of identification. For this purpose, the VMS **110** executes a lookup of a camera identifier for each camera **103** in the camera locations table **21** to obtain the location (e.g. room number) of each camera **103**. The VMS **110** then reports this information in the surveillance system meta data stream, including each identified individual and room and time, to the SIS **80**.

(37) At the same time, in step **218**, the security integration system **80** requests a list of authorized computer systems **179** for each of the employees from the authentication server **22** of the IT system **20**. The authentication server **22** then returns the list of authorized computer systems **179** for each of the employees in step **220**.

(38) The SIS **80** further determines the locations of the various computer systems **122** relative to the rooms **113** from the asset management database **30**. This is indicated in step **222**. For this purpose, the SIS **80** queries the computer locations table **96** of the asset management database **30**. The asset management database **30** returns the computer locations relative to the rooms **113** back to the SIS **80** in step **224**.

(39) Then, based on the list of authorized computer systems for each individual and the computer systems in each room, the SIS **80** provides a white list of valid potential users, i.e., the valid users that are known to be in the same room as the computer systems, for each computer system **122**. The SIS **80** provides the white list to the authentication server **22**, in step **226**.

(40) The computer systems **122** might then receive user login requests from individuals **60** within the rooms, in step **228**. For example, one of the individuals may try to logon to one of the client computers **122** by entering a username and password. The user login requests include at least user credentials (e.g. user ID, login ID) and a computer system name. The computer systems then send the user login requests including the user credentials and the requested computer system names to the authentication system **22** for further processing.

(41) The authentication server **22**, in step **230**, compares the user credentials and the requested computer system names in the login requests to the white list of valid potential users for each computer system **122**. Upon finding a matching entry in the white list for that individual **60**, the authentication system **22** authenticates that individual on that computer system and might further send a message to the computer system **122** to allow access to the computer system **122** named in the user login request. Otherwise, the authentication server **22** denies/blocks access to the computer system named in the user login request when it has been determined that the user is not even present in the room, for example.

(42) FIG. 2B is a sequence diagram that illustrates an alternate implementation for the enterprise security system **100**.

(43) In FIG. 2B, steps **202** through **216** are substantially similar as steps **202** through **216** in FIG. 2A.

(44) In step **280**, the security integration system **80** requests a list of authorized computer systems for each of the employees from the authentication server **22** of the IT system **20**. The authentication server **22** then returns the list of authorized computer systems **179** for each of the employees in step **282**.

(45) The SIS **80** further determines the locations of the various computer systems **122** relative to the rooms **113** from the asset management database **30**. This is indicated in step **284**. For this purpose, the SIS **80** requests the computer locations table **96** of the asset management database **30**. The asset management database **30** returns the computer locations relative to the rooms **113** back to the SIS **80** in step **286**.

(46) Then, based on the list of authorized computer systems for each individual and the computer systems in each room, the SIS **80** provides a black list of denied users for each computer system **122**. In step **288**, the SIS **80** provides the black list for each of the individuals **60** back to the authentication server **22**.

(47) The computer systems **122** then receive user login requests from individuals **60** within the rooms, in step **290**. This is substantially similar to step **288** in FIG. **2A**. The computer systems **122** then send the user login requests including the user credentials and the requested computer system names to the authentication system **22** for further processing.

(48) The authentication system **22**, in step **292**, compares the user credentials and the requested computer system names in the login requests to the black list of denied users for each computer system **122**. Upon finding a matching entry in the black list for that individual **60**, the authentication server **22** blocks access to the computer system **122** named in the user login request.

(49) In this way, the enterprise security system **100** also tracks movement of individuals throughout an enterprise, provides access to computer resources within the enterprise using an authentication server, and blocks access to the computer resources based on the tracked movement of the individuals within the enterprise.

(50) FIG. **3** illustrates another method of operation for the enterprise security system **100**.

(51) As before, the VMS **110** provides a surveillance metadata stream to the SIS **80**. The stream includes identified employees/individuals and location(s) in the building **50** where they were tracked along with a timestamp indicating the time of detection. The ACS **128** provides its access control meta data stream after employees interact with card readers, for example.

(52) Now, the SIS **80** also accesses the ERM system **138** to determine the assigned desks for those employees and then confirms that the employees are on valid paths to or from their desks **45**, in one example.

(53) As before, the enterprise security system **100** includes one or more tracking systems for tracking movement of the individuals **60** throughout the enterprise. In this way, the SIS **80** receives a metadata stream of the tracked movements of the employees and determines whether the employees are moving to and/or from the respective desks of the employees by reference to information from the ERM **138**.

(54) In more detail, in step **250-1**, camera **103-1** sends image data **99** captured of a scene within room **113-1** to the VMS **110**. In a similar vein, camera **103-2** in step **250-2** sends image data **99** captured of a scene within room **113-2** to the VMS **110**.

(55) In step **252**, the facial recognition module **107** of the VMS **110** locates the individuals **60** in the image data **99**. The facial recognition module **107** then performs facial recognition of the individuals located in the image data **99** to identify the individuals **60** in step **254**.

(56) According to step **256**, the VMS **110** sends the metadata stream of information for any employee/individuals **60** identified, where the stream includes a building name, location (e.g. room **113**) within the building **50**, and a time stamp for each identified employee, in examples, to the SIS **80**.

(57) Also, as before, the ACS **128** also preferably provides its access control meta data stream to the SIS **80**.

(58) In step **258**, the SIS **80** requests a list of assigned desk locations **161** for each identified employee. The desk locations **161** are included in the employee records **123** of the individuals, which the ERM system **138** maintains in its employee database **139**. The ERM system **138** sends the list of assigned desk locations **161** for each identified employee back to the SIS **80** in step **260**.

In addition, the ERM system **138** sends the badge photo **36** for each identified employee to the SIS **80** in step **262**.

(59) In step **264**, the SIS **80** determines if each of the identified employees are traversing a path through the building(s) **50** that is consistent with the list of assigned desk locations for each employee.

(60) The SIS **80** also determines, in step **266**, if surveillance images obtained of the rooms (at substantially the same times as the time stamps in the stream) include badge photos **36** that match the identified employee badge photos. A mismatch indicates that the individuals **60** in the rooms **113** may have stolen or forged their security badges. The SIS **80** can then send messages alerting security operators of this event so that the security operators can investigate further.

(61) The method of operation for the enterprise security system **100** in FIG. **3** has additional benefits. The system might be used to monitor usage of desks **45** by the individuals **60** over time. This could then be provided as input to future real estate occupancy decisions by analyzing the space usage over a longer period. For example, if the occupancy is less than 80% of the available desk **45** space, an operator at the enterprise may wish to consider moving to a smaller building **50**. Similarly, if the desk **45** space usage exceeds 90%, then, if the enterprise wishes to expand, it may need to consider moving to a larger building **50**.

(62) While this invention has been particularly shown and described with references to preferred embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the scope of the invention encompassed by the appended claims.

Claims

1. An enterprise security system, comprising: one or more tracking systems for tracking, using facial recognition to identify individuals based on images captured using one or more surveillance cameras installed in multiple rooms of an enterprise building, movement of the individuals throughout the multiple rooms, wherein the one or more tracking systems include an access control system for controlling physical access of the individuals through access points within the enterprise building, a first access point of the access points permitting physical access to a room of the enterprise building and having a reader device that is included in the access control system; an information technology (IT) system for providing access to computer resources within the enterprise building; and a security integration system for one of: receiving, from the IT system, a list of authorized computers for each of the individuals, and providing, to the IT system based on the list of authorized computers, a white list identifying one or more of the individuals that are determined, based on a metadata stream of the tracked movement of the individuals, to be in a same room as the authorized computers; or receiving, from the IT system, a list of unauthorized computers for each of the individuals, and providing, to the IT system based on the list of unauthorized computers, a black list identifying one or more the individuals that are determined, based on the metadata stream of the tracked movement of the individuals, to be in a same room as the unauthorized computers; wherein the metadata stream includes location within the enterprise building and time stamp for each identified individual, wherein the IT system allows or blocks access, requested by the one or more of the individuals, to at least a portion of the computer resources based on the white list or the black list.

2. The system as claimed in claim 1, wherein the security integration system determines at least the portion of the computer resources as authorized computer systems for the one or more of the individuals identified in the white list, and wherein the IT system blocks access to the authorized computer systems, for one of the individuals that is not identified in the white list.

3. The system as claimed in claim 1, wherein the security integration system accesses an asset management database to determine locations of computer systems within the enterprise building.

4. An enterprise security system, comprising: one or more tracking systems for tracking, using facial recognition to identify an individual based on images captured using one or more surveillance cameras installed in multiple rooms of an enterprise building, movement of the individual throughout the multiple rooms, wherein the one or more tracking systems include an access control system for controlling physical access of individuals through access points within the enterprise building, a first access point of the access points permitting physical access to a room of the enterprise building and having a reader device that is included in the access control system; an employee resource management system including an employee database indicating location of a desk of the individual; a security integration system for determining, based on receiving a metadata stream of the tracked movement of the individual, whether the individual is moving to and/or from the desk of the individual, wherein the metadata stream includes location within the enterprise building and time stamp for the individual; providing, based on a list of authorized computer resources, a white list identifying the individual determined, based on the metadata stream, to be moving to the desk of the individual associated with the authorized computer resources; and providing, based on the white list, access for the individual to the authorized computer resources.

5. A method, comprising: tracking, using facial recognition to identify an individual based on images captured using one or more surveillance cameras installed in multiple rooms of an enterprise building, movement of the individual throughout the multiple rooms, wherein tracking movement includes controlling and reporting, via an access control system and to a metadata stream, physical access of individuals through access points within the enterprise building, a first access point of the access points permitting physical access to a room of the enterprise building and having a reader device that is included in the access control system, wherein the metadata stream includes location within the enterprise building and time stamp for the individual; providing, based on a list of authorized computer resources, a white list identifying the individual determined, based on the metadata stream, to be in a same room as the authorized computer resources; providing, based on a list of unauthorized computer resources, a black list identifying the individual determined, based on the metadata stream, to be in a same room as the unauthorized computer resources; providing, based on the white list, access for the individual to the authorized computer resources within the enterprise building using an authentication server; and blocking, based on the black list, access for the individual to the unauthorized computer resources determined, based on the tracked movement of the individual within the enterprise building, to be in a same room as the individual.

6. The method as claimed in claim 5, further comprising instructing an IT system to block access to the authorized computer resources, based on the tracked movement of the individual, when individual is not present.

7. The method as claimed in claim 6, further comprising accessing an asset management database to determine locations of computer systems within the enterprise building.

8. A method, comprising: tracking, using facial recognition to identify an individual based on images captured using one or more surveillance cameras installed in multiple rooms of an enterprise building, movement of the individual throughout the multiple rooms, wherein the tracking movement includes controlling and reporting, via an access control system, physical access of individuals through access points within the enterprise building, a first access point of the access points that permits physical access to an area of the enterprise building and having a reader device that is included in the access control system; receiving a metadata stream of the tracked movement of the individual, wherein the metadata stream includes location within the enterprise building and time stamp for the individual; determining, based on the metadata stream of the tracked movement of the individual, whether the individual is moving to and/or from a desk of the individual based on information from an employee resource management system including an employee database indicating location of the desk of the individual; providing, based on a list of authorized computer resources, a white list identifying the individual determined, based on the

metadata stream, to be moving to the desk of the individual associated with the authorized computer resources; and providing, based on the white list, access for the individual to the authorized computer resources.

9. The method as claimed in claim 8, further comprising determining an assigned desk location of the desk of the individual and comparing the tracked movement to the assigned desk location.

10. The method of claim 8, wherein the controlling the physical access of the individuals through the access points within the enterprise building comprises: receiving, from a reader device installed at the first access point, one or more credentials of the individual proximate to the first access point; and causing, based on the one or more credentials, the first access point to grant physical access to the at least one of the multiple rooms.

11. The method of claim 10, where the first access point is a door, and the causing comprises sending a signal to unlock a door lock of the door.

12. A method, comprising: tracking, by tracking systems, using facial recognition to identify an individual based on images captured using one or more surveillance cameras installed in multiple rooms of an enterprise building, movement of the individual throughout the multiple rooms, wherein the tracking systems include an access control system for controlling physical access of individuals through access points within the enterprise building, a first access point of the access points that permits physical access to a room of the enterprise building and having a reader device that is included in the access control system; obtaining, via an employee resource management system, location of a desk of the individual, the employee resource management system including an employee database indicating the location of desks of the individuals; receiving, by a security integration system, a metadata stream of the tracked movement of the individual, wherein the metadata stream includes location within the enterprise building and time stamp for the individual; determining, based on the metadata stream of the tracked movement of the individual, whether the individual is moving to and/or from the desk of the individual by reference to information stored in the employee database; providing, based on a list of authorized computer resources, a white list identifying the individual determined, based on the metadata stream, to be moving to the desk of the individual associated with the authorized computer resources; and providing, based on the white list, access for the individual to the authorized computer resources.
